Hierarchical GNN Framework for Earth's Surface Anomaly Detection in Single Satellite Imagery

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***Abstract*— Sudden-onset Earth's surface anomalies, such as natural disasters and man-made incidents, pose severe threats to human life and property security, emphasizing the crucial role of accurate detection and rapid response in Humanitarian Assistance and Disaster Response (HADR). In this work, we propose a hierarchical graph neural network (GNN)-based framework for Earth's surface anomaly detection, called L2S-Net, to integrate from local-to-semantic (L2S) information for rapid and accurate detection of multiclass anomalies. Specifically, L2S-Net only uses a single very high-resolution (VHR) image as input to expedite processing speed, while using a hierarchical graph representation for better image understanding. Meanwhile, drawing from brain-inspired research and graph theory, we design an L2S fusion network, called L2S-GNN, to explicitly learn relationships between nodes at different levels facilitating accurate detection of Earth's surface anomaly. L2S-Net significantly reduces data requirements while capturing valuable higher order information from images, achieving a superior balance between accuracy and efficiency. Furthermore, due to the lack of a public dataset for Earth's surface anomaly detection, we create a novel and large-scale benchmark dataset ESADv2. Extensive experiments on the ESADv2 dataset and two real-world cases demonstrate that the proposed L2S-Net outperforms many state-of-the-art methods in both model size and performance while exhibiting exceptional generalizability and robustness.**

***Index Terms*— Disaster response, Earth's surface anomaly, graph neural network (GNN), rapid detection, remote sensing (RS).**

I. Introduction

THE Earth's surface anomaly is "the phenomenon threatening the natural or social environment, which destroys the stable balanced state of the Earth's surface conforming to the historical evolutionary law caused by the single or double influence of natural and human factors" [1], [2]. Over the past decades, Earth's surface anomaly including natural disasters and man-made incidents have become increasingly frequent, resulting in significant human casualties, displacement, and property loss [3], [4]. Therefore, timely anomaly detection and response play a pivotal role in Humanitarian Assistance and Disaster Response (HADR) [4], [5], [6].

In recent years, the development of deep learning (DL) has led to significant progress in remote sensing (RS)-based HADR tasks. Nevertheless, there are still two primary challenges. The first challenge revolves around timeliness. Current methods mostly follow a post hoc analysis paradigm. Multitemporal methods, primarily using change detection, excel in accurately localizing the affected areas [7], [8], [9]. Meanwhile, multimodal methods are also well-studied, which incorporate additional modalities such as infrared and radar data to capture valuable information not present in very high-resolution (VHR) images [10], [11], [12], [13]. However, acquiring and processing such additional data is time-consuming or even unavailable, thereby hindering HADR-related tasks that require as low latency as possible [14], [15], [16]. For rapid detection and response, some recent studies have achieved remarkable strides, by deploying lightweight models on resource-constrained platforms such as drones, satellites, and significantly reducing the time of data transmission and processing [6], [15], [17].

However, all these methods face the same second challenge, namely, the limitation of the dataset, both in terms of anomaly classes and the scale of the dataset. Many current studies focus only on single or a few anomalies, using relatively small datasets [11], [13], [18], [19]. While these endeavors have achieved outstanding performance, in the real world, training ad hoc models tailored to each sudden anomaly is labor-intensive and inefficient [20]. Some studies have released larger HADR-related datasets, such as xBD [5] and ADSF [20], for training more robust and generalizable models. However, to the best of our knowledge, there is currently no large-scale dataset designed for multiclass anomalies' detection. In the context of data scarcity, several studies attempt to address this problem through unsupervised learning methods. However, the extensive and intricate information of satellite imagery leads to suboptimal performance [21], [22].

To address the aforementioned challenges, we propose a graph neural network (GNN)-based framework, called

Manuscript received 11 March 2024; revised 2 May 2024; accepted 28 May 2024. Date of publication 3 June 2024; date of current version 13 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42192580 and Grant 42192583, in part by Hubei Province Natural Science Foundation under Grant 2021CFA088, in part by the Civilian Space Project under Grant D040103 and Grant D010206, and in part by Wuhan University-Huawei Geoinformatics Innovation Laboratory. *(Corresponding author: Zhi Gao.)*

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Digital Object Identifier 10.1109/TGRS.2024.3408330

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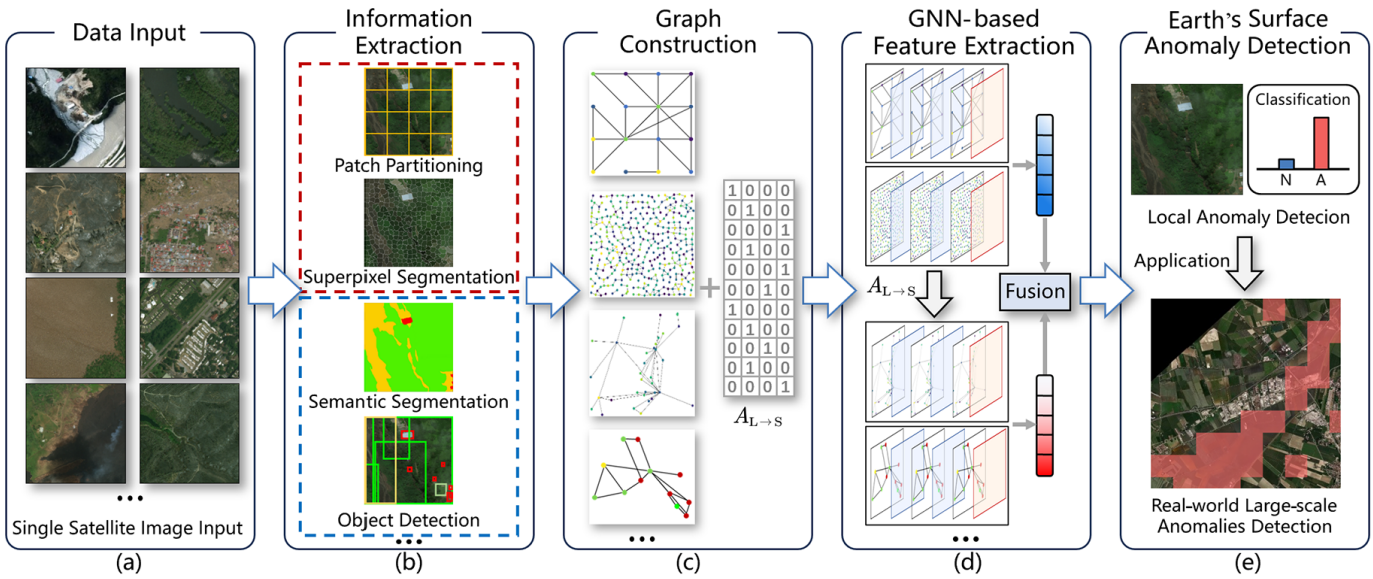

Fig. 1. Overview of L2S-Net framework that consists of five steps. (a) Inputs a single satellite image. (b) Information extraction: extracts local objects (red dashed box) and geoentities (blue dashed box). (c) Graph construction: constructs graphs from extracted objects as nodes and forms a hierarchical graph structure based on intergraph relationships. (d) GNN-based feature extraction: uses a well-designed GNN to extract the fused graph embedding. (e) Earth's surface anomaly detection: uses a binary classifier for rapid anomaly detection, and the entire process can be scalable to large-scale real-world applications.

L2S-Net, which integrates from local-to-semantic (L2S) information for rapid and accurate Earth's surface anomaly detection. As illustrated in Fig. 1, L2S-Net consists of five key steps. To tackle the first challenge, we only use a single VHR satellite image as input [Fig. 1(a)], thus mitigating the time delay of data collection and processing. Furthermore, we adopt an image-level binary classification manner to facilitate timely detection and response, for preserving fewer samples for efficient downstream tasks. In response to the second challenge, we have created a novel, large-scale dataset called ESADv2 (Earth's surface anomaly detection) for multiclass anomalies' detection. Specifically, inspired by visual perception theory and brain-inspired research [23], [24], [25], which highlights the use of visual characteristics and relationships for better image understanding, L2S-Net first constructs a hierarchical graph representation to integrate from L2S information as shown in Fig. 1(b) and (c). After that, L2S-Net uses a tailored GNN-based network called L2S-GNN [Fig. 1(d)] to leverage the semantics and relationships of ground objects at different levels to extract the graph embedding for final anomaly detection as shown in Fig. 1(e).

Our main contributions could be summarized as follows.

1) We propose L2S-Net, a framework for rapid and accurate detection of Earth's surface anomaly in single satellite imagery. To the best of our knowledge, our work is the first attempt of GNN for RS Earth's surface anomaly detection.
2) We propose a hierarchical graph representation for RS imagery and design the L2S-GNN to explicitly guide the model in efficiently learning hierarchical visual characteristics and semantic relationships among ground objects.
3) We create a novel large-scale dataset ESADv2 for Earth's surface anomaly detection and conduct extensive experiments on ESADv2 and two large-scale real-world cases to demonstrate that L2S-Net outperforms current advanced methods. A series of ablation studies are conducted to evaluate the effectiveness of each component.

The rest of this article is organized as follows. Section II provides a review of related works. Section III describes the details of proposed L2S-Net. Section IV presents extensive experiments results and discussion, and the conclusion is summarized in Section V.

II. Related Works

In this section, we review the works closely related to this study, including Earth's surface anomaly in RS, GNN, and the integration of GNN in RS.

A. Earth's Surface Anomaly in RS

RS plays a crucial role at each stage of intelligent analysis of Earth's surface anomaly, including the preparedness, mitigation, response, and recovery phases [26]. Recent research in RS-based methods of Earth's surface anomaly mainly uses a DL-based model, particularly focusing on post hoc mapping and damage assessment. Change detection has emerged as a widely studied method, such as landslide area extraction [9], and building damage assessment [7], [8]. For specific anomaly, multimodal methods have demonstrated outstanding performance. For instance, SAR images have been introduced for landslides and flooding detection [10], [12], while infrared images have been used for coal fire detection [11]. However, these methods bear the burden of time for data transmission and processing and even become impractical when additional data are unavailable. To enable faster response times, models such as lightweight EmergencyNet [17] and flooding segmentation model in [6] have been proposed. Some models have even been deployed on

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satellites to facilitate real-time processing and thereby reduce the latency [15].

Despite significant efficiency improvements, the prevalent issue remains: current methods typically rely on labeled datasets [27]. Unfortunately, most of the related datasets are designed for limited classes of anomaly or specific case studies, making it challenging for models to generalize across different anomalies and regions in real-world. Consequently, inspired by the success of unsupervised learning in anomaly detection, several studies have explored strategies such as style transfer and image generation [22], [28]. However, compared to images used in unsupervised anomaly detection, satellite imagery is far more complicated, resulting in unsatisfying results for these methods. In a recent breakthrough, a global prior knowledge database was proposed for rapid and accurate anomaly detection of specific anomaly classes [2]. While promising, this approach is still in the early stages of practical application and necessitates further investigation.

In short, there is still a lack of a solution that enables the rapid and comprehensive detection of diverse Earth's surface anomalies. In this article, we propose L2S-Net, a novel framework designed for rapid and accurate detection of Earth's surface anomaly using only a single VHR image. Meanwhile, we create a more comprehensive and large-scale benchmark dataset ESADv2, to tackle the problems of model training.

B. Graph Neural Networks

In the past few years, GNNs have gained increasing attention. In contrast to traditional DL-based methods that implement on pixels or patches [29], [30], GNNs are more flexible and able to handle irregular inputs and preserve the valuable semantic information. Moreover, GNNs can explicitly learn relationships among entities, which is beneficial for image understanding from higher level. The powerful representational capability of graph enables GNNs to naturally perform tasks such as node classification, link prediction, and graph classification [31]. Due to the characteristics mentioned above, GNNs have been incorporated into various visual tasks. For instance, ViG constructs patches into graphs for model training and outperforms advanced CNN, MLP, and transformer models in ImageNet [32]. Relation Network [33] effectively enhances object detection and duplicate removal by modeling relationships among object attributes such as appearance and geometry based on GNNs. GloRe introduces GNNs for relational reasoning and obtains state-of-the-art performance in semantic segmentation [34]. A-LCN+A-NLSOA models the image scene as a knowledge graph, using a knowledge-based attention mechanism to focus on relevant entities and knowledge, thereby achieving superior visual question answering [35].

In light of these advantages, we introduce GNNs to capture relationships among ground objects in RS scene. Meanwhile, we observed that GNNs have great data compression capabilities and are known for their lightweight structure, significantly enhancing model inference efficiency, which is highly beneficial for rapid Earth's surface anomaly detection.

C. GNNs in RS

Due to the advantages of GNNs and their powerful modeling capabilities suited for RS tasks, GNNs have been introduced into research in the RS field, particularly focusing on hyperspectral image (HSI) tasks. CEGCN, which combines CNN and GNN, uses spectral features to model spatial and semantic relationships, thereby improving the accuracy of land cover mapping [36]. By constructing different multiscale graphs, many GNN-based methods have notably improved the performance of semantic segmentation of HSI [37], [38], [39]. GiG achieves a superior integration of low-level pixel features and high-level geometric features, thereby effectively capturing relationships within and between superpixels [40]. In the field of HSI anomaly detection, GNNs are also a widely used method [41], [42], [43]. In the aforementioned methods, segmented superpixels are commonly used for constructing graphs. However, while superpixels facilitate data abstraction and representation, they also result in the loss of valuable semantic information. Instead, several studies have noted that explicitly embedding semantics into nodes is beneficial to model performance. H-LCN uses deep features as node attributes to represent ground objects and achieve accurate complex scene classification [44]. MLRSSC-CNN-LNN uses an object detection network to extract objects as graph nodes and achieves high-accuracy image retrieval [45].

Overall, GNNs are of great promise in the RS field. Moreover, it is observed that different graph construction strategies significantly influence GNN performance. Local entities, such as superpixels, are rich in low-level details yet lack semantic information. Conversely, semantic objects obtained through strategies such as semantic segmentation and object detection exhibit the opposite characteristics. In this article, L2S-Net transforms a single VHR image into a hierarchical graph, and L2S-GNN is proposed to leverage the strengths of both local and semantic information.

III. Our Method

In this section, we instantiate L2S-Net for a detailed description of the proposed framework, including hierarchical graph construction and the details of L2S-GNN.

A. Notation and Preliminary

The L2S-Net adopts a supervised learning manner for image-level binary classification, categorizing various anomalies, such as earthquakes and floods, within the anomaly class. The setting facilitates the rapid filtering of a large number of normal samples for downstream ad hoc applications. Given a single image $\mathcal{X}$ input, the objective is to rapidly categorize it into a binary label $\mathcal{Y} \in (0, 1)$, where 0 denotes normal and 1 indicates anomaly. As illustrated in Fig. 1, $\mathcal{X}$ is initially transformed into an attributed graph $G = (V, E, \mathbf{X})$ through information extraction and graph construction. Here, V and E represent sets of nodes and edges, respectively, with $v \in V$ denoting a single node, and $e_{vu} = (v, u) \in E$ denoting an edge from node v to node u. $\mathbf{X} \in \mathbb{R}^{n \times d}$ represents the matrix of all the node features, $n = |V|$ represents the number of nodes, $|\cdot|$ is the length of the set, and d represents the length

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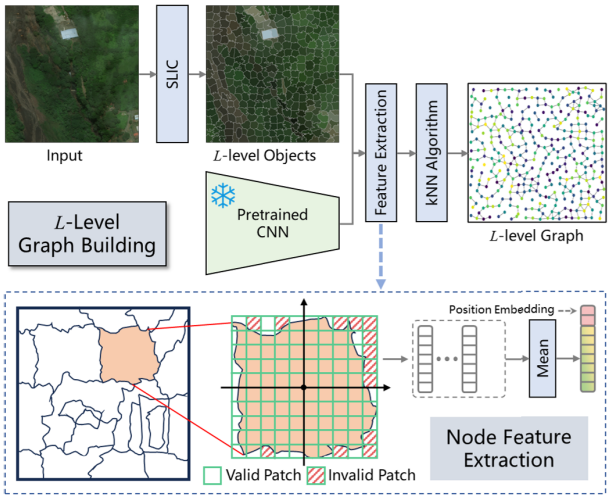

Fig. 2. Process of L-level graph construction. The input single satellite imagery is used to generate the L-level objects by SLIC algorithm as nodes, the features of each node are extracted by sliding window using pretrained CNN, and finally the topology of the graph is established by the kNN algorithm.

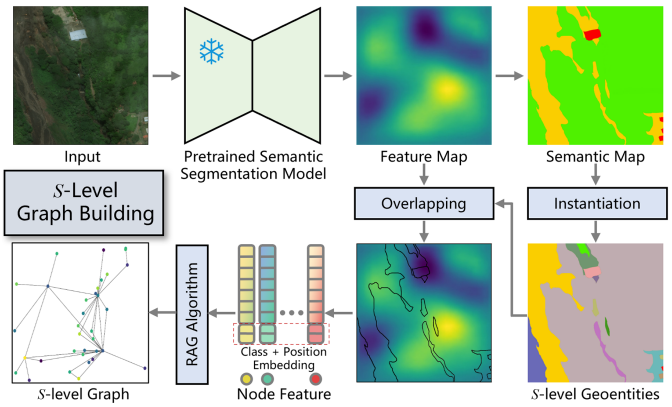

Fig. 3. Process of S-level graph construction. The input single satellite imagery is input to a pretrained semantic segmentation model to generate the segmentation result and middle feature map. Each connected component is treated as a node, with its features jointly represented by the mean vector of the feature map combined with class and position embeddings. The graph topology is then established based on using the RAG algorithm.

of the node feature. For any node v, its neighborhood can be represented as $N(v) = \{u \in V \mid (v, u) \in E\}$. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is used to describe the topology of G, where $\mathbf{A}_{ij} = 1$ if $e_{ij} \in E$ and $\mathbf{A}_{ij} = 0$ otherwise. G and $\mathbf{A}$ are fed into a GNN for learning and updating, generating the graph embedding for subsequent anomaly detection.

B. Hierarchical Graph Construction

For the instantiation of the L2S-Net, after considering factors such as feasibility of the method and model size, we use superpixels and semantic segmentation for graph construction, denoted as L(local)-level graphs and S(semantic)-level graphs, corresponding to the red and blue dashed boxes in Fig. 1(b). Simultaneously, to fully leverage the semantic relationships between graphs, we introduce a distribution matrix to form a hierarchical graph representation, corresponding to Fig. 1(c).

1) L-Level Graph: The process of L-level graphs' building is illustrated in Fig. 2. Following common practices in previous works, we use the simple linear iterative clustering (SLIC) algorithm [46] to obtain superpixels, denoted as L-level objects $\boldsymbol{R}_{\mathrm{L}}$. Each object $r_{\mathrm{L}} \in \boldsymbol{R}_{\mathrm{L}}$ exhibits relatively homogeneous appearance and can be approximated by the same feature representations, thus serving as nodes.

Subsequently, we use feature vectors extracted by a pretrained CNN $f_{\mathrm{L}}^{\mathrm{pre}}(\cdot)$, as node attributes. It is noteworthy that the CNN weights are frozen, not updated during model training. Specifically, for better utilization of information from each superpixel, we use a sliding patch with a fixed width W around the geometric centroid for feature extraction. For each patch p, we compute its effective pixel ratio r_p as follows:

$$r_p = \frac{1}{W^2} \sum_{i=1}^{W} \sum_{j=1}^{W} \mathbb{I}((i, j) \in p), \; p \in r_{\mathrm{L}} \tag{1}$$

where (i, j) is the pixel in p, and $\mathbb{I}(\cdot)$ is the indicator function.

We introduce a threshold τ, set to 0.5 in our experiments, to determine the validity of p. Each p will be resized and input into $f_{\mathrm{L}}^{\mathrm{pre}}(\cdot)$ to extract the feature vector when $r_p > \tau$. The average feature vector of all the valid patches is concatenated with the position embedding of the current r_{L} to form the node attribute. This process can be formulated as follows:

$$\mathrm{x}_{\mathrm{L}}(v) = \left[\frac{1}{m_1} \sum_{i=1}^{m_1} f_{\mathrm{L}}^{\mathrm{pre}}(\mathrm{RESIZE}(p_i)) \middle\| \mathbf{E}_{\mathrm{pos}} \right] \tag{2}$$

where $\mathrm{x}_{\mathrm{L}}(v) \in \mathbb{R}^{d_1}$ is the feature vector of the node v, and m_1 is the number of valid patches of current r_{L}. RESIZE$(\cdot)$ is the bilinear interpolation operation. $\mathbf{E}_{\mathrm{pos}}$ is the position embedding of r_{L}, and $\|$ is the concatenation operator.

We use the k-nearest neighbors (kNN) algorithm to establish graph by linking nodes based on the distance between their features. However, imposing a strict limitation on the value of k can lead to the connection of some unrelated nodes, introducing additional noise. Therefore, we set a distance threshold δ to prune the initial graph to obtain the final L-level graph $G_{\mathrm{L}} = (V_{\mathrm{L}}, E_{\mathrm{L}}, \mathbf{X}_{\mathrm{L}})$. In our experiments, we set $k = 5$ and $\delta = 60$. For any node v, neighbors u satisfying the following condition will be connected by e_{vu}:

$$E(v, u) = \begin{cases} 1, & \text{if } d(v, u) < \delta \wedge u \in N(v, k) \\ 0, & \text{else} \end{cases} \qquad \forall v, u \in V_{\mathrm{L}} \tag{3}$$

where $d(v, u)$ denotes the distance between node v and u, $N(v, k)$ is the set of kNNs of node v, and V_{L} means the node set corresponding to the current L-level graph.

2) S-Level Graph: The building of S-level graphs is illustrated in Fig. 3. S-level graph centers on the semantics of ground objects, constructing a graph by explicitly linking semantic regions and guiding the model to learn useful higher order relationships (e.g., cars should be on roads, not submerged in water). Drawing on study [47] on RS scene graph generation via semantic segmentation, we use a pretrained model for extraction of semantic regions and

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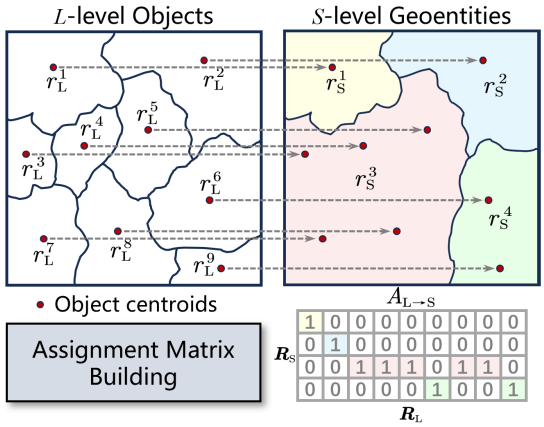

Fig. 4. Process of assignment matrix building. The geometric centroid of each r_{L}^{i} is spatially mapped to $\boldsymbol{R}_{\mathrm{S}}$, and a binary assignment matrix $A_{\mathrm{L}\to\mathrm{S}}$ is constructed based on spatial location. Here, different colors in $\boldsymbol{R}_{\mathrm{S}}$ represent distinct semantic regions, where the number 1 in $A_{\mathrm{L}\to\mathrm{S}}$ indicates the association of r_{L}^{i} with r_{S}^{j}, while 0 denotes no correlation.

intermediate feature map. Similarly, the model weights remain frozen during the training stage. We instantiate the semantic segmentation results to obtain all the connected components, denoted as S-level geoentities $\boldsymbol{R}_{\mathrm{S}}$, with each geoentity $r_{\mathrm{S}} \in \boldsymbol{R}_{\mathrm{S}}$ acting as a node. After that, we superimpose the feature maps with $\boldsymbol{R}_{\mathrm{S}}$ and compute the average feature vector for each r_{S} to represent the node attributes, formulated as follows:

$$\mathrm{x}_{\mathrm{S}}(v) = \left[\frac{1}{m_2}\sum_{i}^{m_2}\mathbf{p}_i \,\middle\|\, \mathbf{E}_{\mathrm{pos}} \,\middle\|\, \mathbf{E}_{\mathrm{class}}\right] \tag{4}$$

where $\mathrm{x}_{\mathrm{S}}(v) \in \mathbb{R}^{d_2}$ is the feature vector of the node v. m_2 is the number of pixels of the current r_{S}, and $\mathbf{p}_i$ represents the feature vector of each pixel in the feature map. $\mathbf{E}_{\mathrm{class}}$ is the semantic embedding of r_{S} through one-hot encoding.

Leveraging the prevalent semantic relationships among adjacent geoentities, we use the region adjacency graph (RAG) algorithm for node connection, thus constructing the S-level graph $G_{\mathrm{S}} = (V_{\mathrm{S}}, E_{\mathrm{S}}, \mathbf{X}_{\mathrm{S}})$.

3) Assignment Matrix: For a large semantic region, such as expansive forest, the node attribute represented with an average feature vector may result in the loss of valuable low-level information. To more effectively learn visual features and semantic relationships among ground objects, we introduce an assignment matrix $A_{\mathrm{L}\to\mathrm{S}} \in \mathbb{R}^{|\boldsymbol{R}_{\mathrm{L}}|\times|\boldsymbol{R}_{\mathrm{S}}|}$ to construct a hierarchical graph to integrate local and semantic information. Specifically, as illustrated in Fig. 4, each geometric centroid θ_{L}^{i} of r_{L}^{i} is mapped onto the $\boldsymbol{R}_{\mathrm{S}}$ based on spatial distribution. The binary matrix $A_{\mathrm{L}\to\mathrm{S}}$ will be established according to the relative spatial relationships between r_{L} and r_{S}, following the rules:

$$A_{\mathrm{L}\to\mathrm{S}}[i,j] = \begin{cases} 1, & \text{if } \theta_{\mathrm{L}}^{i} \in r_{\mathrm{S}}^{j} \\ 0, & \text{else.} \end{cases} \tag{5}$$

In $A_{\mathrm{L}\to\mathrm{S}}$, each r_{L} is exclusively associated with one and only one r_{S}. Particularly, the overlap between each r_{L} and adjacent r_{S} will be calculated when it is at the border of multiple r_{S}, and it will be assigned to r_{S} with the maximum overlap. Consequently, the local-semantic joint representation of the hierarchical graph is constructed as $G_{\mathrm{L2S}} = \{G_{\mathrm{L}}, G_{\mathrm{S}}, A_{\mathrm{L}\to\mathrm{S}}\}$.

C. L2S-GNN

For learning and updating the hierarchical graph G_{L2S}, we propose a novel architecture named L2S fusion GNN (L2S-GNN), as illustrated in Fig. 5. In brief, two distinct GNNs (the L-level GNN and S-level GNN) are initially used to extract the corresponding graph embeddings based on G_{L2S}, with $A_{\mathrm{L}\to\mathrm{S}}$ guiding the node updates across different GNNs. Finally, an attention-based fusion module (AFM) combined with a classification head is used for integrating graph embeddings and detecting Earth's surface anomaly. It is noted that the entire L2S-GNN is trainable.

1) Feature Update of L-Level Graph: Given the different relationships among ground objects, it is unfair to apply the same weight to all the neighboring nodes during feature aggregation. Therefore, L-level GNN first uses a simple GNN with shallow GATv2 [48] layers to update the features of the nodes in G_{L} as follows to obtain G'_{L}:

$$h^{(l)}(v) = \sigma\left(\sum_{u\in N(v)} \alpha_{vu}^{(l)} W^{(l)} h^{(l-1)}(u) + b^{(l)}\right) \tag{6}$$

where $h^{(l)}(v)$ represents the hidden feature of node v updated after the lth GATv2 layer. $h^{(0)}(v) = \mathbf{x}_{\mathrm{L}}(v)$ if the current input is the initial G_{L}. $W \in \mathbb{R}^{d_1\times d_1}$ and $b \in \mathbb{R}^{d_1}$ are learnable weights and bias terms, respectively, and σ is the sigmoid activation function. $\alpha_{vu}^{(l)}$ represents the attention between nodes v and u, which can be computed as follows:

$$\alpha_{vu} = \frac{\exp(\boldsymbol{\rho}(h(v), h(u)))}{\sum_{k\in N(v)} \exp(\boldsymbol{\rho}(h(v), h(k)))}. \tag{7}$$

In (7), a scoring function $\boldsymbol{\rho} : \mathbb{R}^{d_1} \times \mathbb{R}^{d_1} \to \mathbb{R}$ is used to assign a score to each edge e_{vu} to represent the correlation between the nodes v and u, and a softmax function is applied to calculate the attention. Specifically, the scoring function is computed as follows:

$$\boldsymbol{\rho}(h(v), h(u)) = a^{\top}\mathrm{LeakyReLU}([\,Wh(v) \| Wh(u)]) \tag{8}$$

where $W \in \mathbb{R}^{d_1\times d_1}$ and $a \in \mathbb{R}^{2d_1}$ are learnable terms.

2) Feature Reinitialization of S-Level Graph: It is crucial to supplement the low-level information lost during the construction of G_{S}. Therefore, S-level GNN first uses an assignment module to integrate the features of G'_{L} into G_{S} based on $A_{\mathrm{L}\to\mathrm{S}}$ to obtain G'_{S}. The reinitialization process for any node feature $h_{\mathrm{S}}^{(0)}(v)$ is formulated as follows:

$$h'_{\mathrm{S}}(v) = \left[h_{\mathrm{S}}^{(0)}(v) \,\middle\|\, \sum_{u\in\mathcal{M}(v)} h'_{L}(u)\right] \tag{9}$$

where $\mathcal{M}(v) = \{u \in V_{\mathrm{L}} \mid A_{\mathrm{L}\to\mathrm{S}}[u, v] = 1\}$ is a set of nodes in G_{L}, facilitating the mapping of node features $h'_{L}(u)$ from G'_{L} to the corresponding nodes $v \in V_{\mathrm{S}}$.

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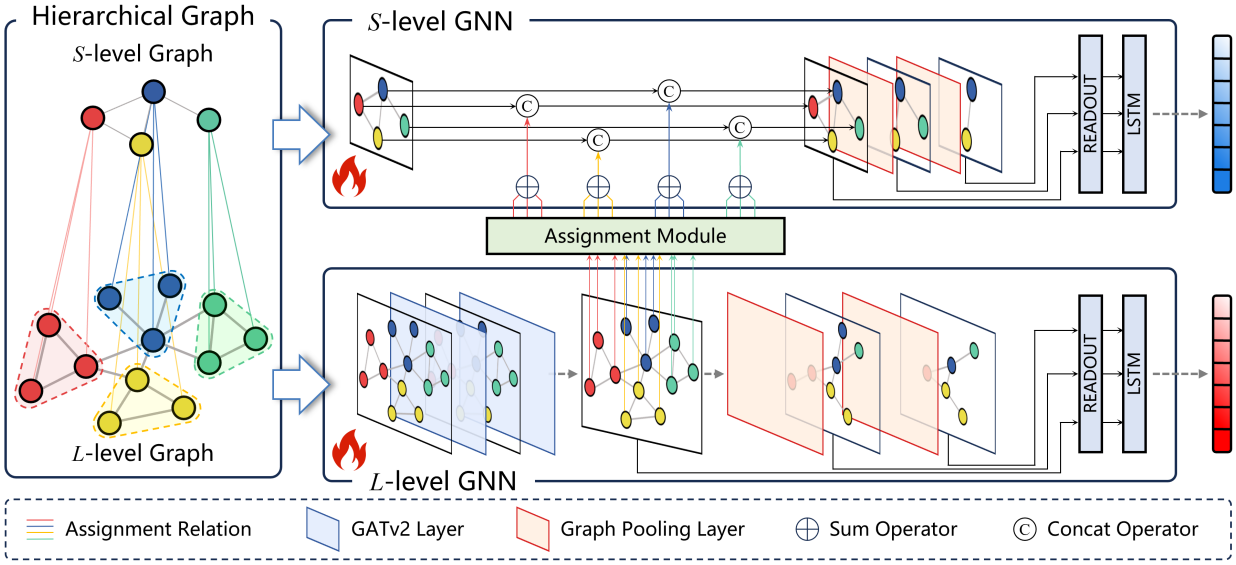

Fig. 5. Overview of L2S-GNN. In the L-level GNN, G_{L} is first updated via a shallow GAT network to obtain G'_{L}, followed by the reinitialization of G_{S} using the assignment module based on G'_{L}. Subsequently, graph embeddings from different layers are obtained through a graph pooling network and then fed into an LSTM module to generate final graph embeddings of L-level GNN and S-level GNN.

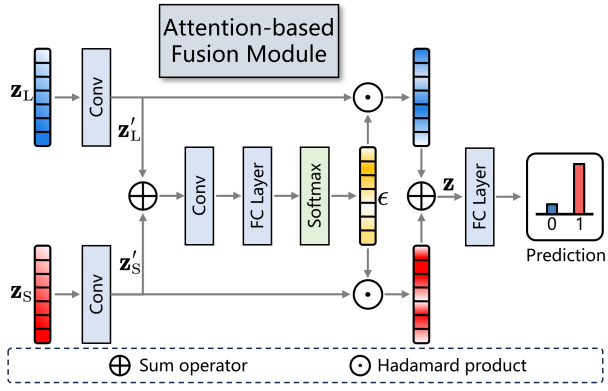

Fig. 6. Overview of AFM. An attention vector ϵ is computed based on inputs $\mathbf{z}_{\mathrm{L}}$ and $\mathbf{z}_{\mathrm{S}}$. Then, graph embeddings are adaptively fused to generate the final feature vector, which is fed into a fully connected layer for the final prediction.

3) GNN-Based Feature Extraction: Inspired by advanced graph classification methods [49], we introduce graph pooling and feature fusion strategies within the L2S-GNN. Specifically, G'_{L} and G'_{S} are first processed through a shallow graph pooling network, enabling node feature updating and graph pruning that can make the entire framework more lightweight. Here, we introduce hierarchical graph pooling with structure learning (HGP-SL), which can adaptively select subsets of nodes for the induced graphs in subsequent layers based on a scoring mechanism, while maintaining the integrity of the topology as much as possible. For more details, the readers are referred to [50].

Drawing on the integration of graph representations from different layers in methods such as those detailed in [50] and [49], we first perform a readout operation on the graphs generated by each graph pooling layer. Subsequently, we use a jump knowledge technique based on LSTM [51] to aggregate the graph embeddings obtained in the previous step, thereby enhancing the expressive power of the final graph embedding. The above process is as follows:

$$\mathbf{r}^{(l)} = \sigma\left[\frac{1}{n^{(l)}}\sum_{p=1}^{n^{(l)}}\mathbf{H}^{(l)}(p,:)\,\Big\|\,\max_{q=1}^{d}\mathbf{H}^{(l)}(:,q)\right] \tag{10}$$

where $\mathbf{r}^{(l)} \in \mathbb{R}^{2d}$ is the graph embedding of the lth layer through readout operation, which involves concatenation and activation of the results of average pooling and max pooling. $\mathbf{H}^{(l)} \in \mathbb{R}^{n^{(l)}\times d}$ is the node features matrix after graph pooling, and $n^{(l)}$ represents the number of nodes.

Then, the graph embeddings from each layer are fed into an LSTM module. Assuming the HGP-SL network comprises k layers, the final graph embedding can be computed as follows:

$$\mathbf{x} = \mathrm{LSTM}\left(\left\{\mathbf{r}^{(t)} \mid t = 0, 1, \ldots, k\right\}\right). \tag{11}$$

To leverage the obtained graph embeddings $\mathbf{x}_{\mathrm{L}}$ and $\mathbf{x}_{\mathrm{S}}$, we draw inspiration from SK-Net [52] and design an AFM (Fig. 6) to adaptively fuse graph embeddings for the final anomaly detection task. The AFM module can be formulated as follows:

$$\begin{aligned}\boldsymbol{\epsilon} &= \mathrm{softmax}\left(\mathrm{FC}\left(\mathrm{CONV}\left(\mathbf{x}'_{\mathrm{L}} + \mathbf{x}'_{\mathrm{S}}\right)\right)\right)\\ \mathbf{z} &= \boldsymbol{\epsilon}\odot\mathbf{x}'_{\mathrm{L}} + \boldsymbol{\epsilon}\odot\mathbf{x}'_{\mathrm{S}}\\ \mathrm{y} &= \mathrm{softmax}(\mathrm{FC}(\mathbf{z}))\end{aligned} \tag{12}$$

where $\mathrm{CONV}(\cdot)$ and $\mathrm{FC}(\cdot)$ represent convolutional and fully connected layers, respectively. $\mathbf{x}'_{\mathrm{L}} = \mathrm{CONV}(\mathbf{x}_{\mathrm{L}})$, as does $\mathbf{x}'_{\mathrm{S}}$. $\mathbf{z} \in \mathbb{R}^{2d}$ serves as the final feature vector, and $\boldsymbol{\epsilon} \in \mathbb{R}^{d}$ is the attention vector used for feature fusion. Finally, in a subsequent step, the feature vector is mapped to its corresponding anomaly class through a fully connected layer followed by a softmax function.

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TABLE I
STATISTICS OF ESADV2

Class		Number of Samples
Anomaly	Flood	647
	Landslide	109
	Debrisflow	48
	Hurricane	1296
	Wildfire	1201
	Earthquake	98
	Volcano	217
	Tornado	254
	Tsunami	160
	Fire	1548
	Bushfire	996
Normal		6821

Fig. 7. Samples of the ESADv2 dataset. Three examples of each class are shown. Normal class are marked in blue, while anomalies are marked in red.

IV. EXPERIMENTS

A. Dataset and Study Area

1) ESADv2 Dataset: The DL-based Earth's surface anomaly detection faces the challenge of scarce labeled datasets. Although some influential HADR-related datasets such as xBD [5] have been released, large-scale datasets specifically tailored for image-level anomaly detection are still lacking. Therefore, we created the ESADv2 dataset to address this gap. To better facilitate Earth's surface anomaly detection in single satellite imagery, we have considered factors such as spatial resolution, coverage area, and ease of acquisition. It integrates some publicly available and redistributable high-quality datasets, augmented with manual annotations, including xBD [5], Multi^{3}Net [53], the Sichuan landslide and debris flow dataset [54], GVLM [55], and the destruction dataset [3], with the majority of the data being VHR images, and a minority being aerial imagery.

In particular, all the samples of ESADv2 were uniformly resized to 1024 × 1024 pixels through operations such as cropping and resizing, with spatial resolutions ranging from 0.2 to 1 m. In such a setting, a single satellite image can cover approximately 4–100 hectares, effectively capturing information on all types of anomalies in the ESADv2 dataset. Each sample was filtered through visual interpretation and labeled as either normal or abnormal, with the latter consisting of 11 classes of anomaly such as floods, landslides, and hurricanes, as illustrated in Fig. 7. Table I details the statistics of ESADv2, which comprises a total of 13 395 samples, including 6821 normal samples and 6574 anomaly samples. It is worth noting that ESADv2 exhibits a significant long-tailed distribution, which not only poses greater challenges for model learning but also reflects the occurrence of various anomalies and public concern in the real world to some extent. Following common practice, the dataset was manually split into training, validation, and test sets in a 6:2:2 ratio.

2) Real-World Case Studies: To further validate the robustness and generalizability of the proposed L2S-Net, we selected two representative real-world cases that were not covered in the ESADv2 dataset: the Beirut port explosion and the Emilia-Romagna flooding in Italy, as shown in Figs. 8 and 9, respectively. These images were downloaded from the Maxar/DigitalGlobe Open Data Program.

On May 2, 2023, the Emilia-Romagna region experienced an unprecedented rainfall that led to severe flooding, inundating dozens of towns and resulting in the evacuation of thousands of residents and numerous injuries. We selected a severely affected subregion as the validation area, covering approximately 155 km^2 with 60 500 × 15 334. This case was used to validate the robustness of the proposed method.

On August 4, 2020, a devastating explosion occurred at the port of Beirut, resulting in widespread destruction and thousands of casualties. We used the administrative divisions of Beirut as the validation area, covering approximately 31.5 km^2 with 13348 × 9466 pixels and a spatial resolution of about 0.5 m. Notably, the explosion is a man-made incident and a not previously seen class out of ESADv2, serving as an excellent test for the generalizability of the proposed method.

B. Implementation Details

All the methods in this article were trained using the Adam optimizer and a plateau learning rate scheduler for 200 epochs. The learning rate was uniformly set to 5e−5, with a weight decay of 1e−5. The scheduler's patience was set to 10, with a decay factor of 0.5. All the experiments were conducted on Ubuntu 18.04 using PyTorch and trained on an NVIDIA 3090 GPU with 24 GB of memory. The batch size for all the methods was set to 128.

In the instantiation of L2S-Net as in Section III, during graph construction, we set the number of superpixels for the SLIC algorithm to 400, a compactness level of ten, a maximum of ten clustering iterations, and a blur kernel size of 1. For the pretrained models used, $f_{\mathrm{L}}^{\mathrm{pre}}(\cdot)$ of L-level GNN is a ResNet-34 pretrained on ImageNet, while $f_{\mathrm{S}}^{\mathrm{pre}}(\cdot)$ of S-level GNN is an HRNetV2 [56] pretrained on the DeepGlobe Landcover Classification dataset [57] and its mIOU is 68.9. The input VHR image will be labeled pixel by pixel into seven categories: urban land, agriculture land, rangeland, forest land, water, barren land, and unknown. Notably, images are resized to 256 × 256 before being input into HRNetV2. For the L2S-GNN, the node feature of L-level graph update uses three GATv2 layers, and both L-level-GNN and S-level-GNN use a three-layer HGP-SL with a pooling rate of 0.5.

Considering our objective is an image-level binary classification task, we use standard cross-entropy loss to optimize

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Fig. 8. Image of Emilia-Romagna flooding.

Fig. 9. Image of Beirut port explosion.

L2S-GNN, and the loss can be calculated as follows:

$$L(y,\hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]. \quad (13)$$

Moreover, we use overall accuracy (OA) and recall rate as evaluation metrics, with their formulas as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \quad (14)$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \quad (15)$$

where TP, TN, FP, and FN are the numbers of samples for true positive, true negative, false positive, and false negative, respectively.

Meanwhile, to demonstrate the advantages of the proposed L2S-Net in model size and inference speed, we compare trainable parameters and FLOPs for further evaluation.

C. Baseline Methods

To more comprehensively evaluate the proposed L2S-Net, we conducted comparisons with many advanced methods. The baseline models include visual baselines and graph-based baselines. Given our target requires both accuracy and efficiency, we selected classic visual models and the latest lightweight image classification models. Moreover, given that GNN is the backbone structure of L2S-Net, we also compared it with several advanced GNNs specifically designed for graph classification tasks.

1) Visual Baselines: ResNet [58] and ViT [59] are the representative works of CNNs and transformer in the computer vision community, respectively. MobileNetv3 [60] and EfficientNet [61] are renowned lightweight CNN models, known for their computational efficiency and performance. Recent advancements in lightweight models such as MobileViT [62], EfficientFormer [63], MobileFormer [64], and EMO [65] have introduced more model design strategies, achieving a good balance between model size and performance.

2) GNN-Based Baselines: HGP-SL [50], Graph U-nets [66], and MEWISPool [67] represent the cutting-edge methods in graph classification tasks, achieving state-of-the-art performance on datasets such as PROTEINS and IMDb. HACT-Net [68] is a model in medical image diagnosis that constructs a hierarchical graph representation from the cell to the tissue level, similar to the graph construction strategy of L2S-Net, thus serving as an important baseline. ViG [32] integrates features of transformers and GNNs, segmenting images into small patches to construct a graph, and achieves the state-of-the-art performance in image classification.

D. Experimental Results

1) Quantitative Results: Earth's surface anomaly detection requires models to be both highly accurate and lightweight, necessitating a great balance between model size and performance. In particular, a high recall rate indicates that the model has a significant advantage in preventing the omission of anomaly detection. Table II demonstrates that the proposed L2S-Net achieves the best performance with an OA and recall rate of 98.20% and 98.19%, respectively, outperforming the classical and larger visual models such as ResNet and ViT. Compared with advanced lightweight visual models such as EMO and MobileFormer, L2S-Net has fewer trainable parameters and comparable FLOPs, yet exhibits superior performance, achieving a better balance between performance and model size. Moreover, compared with GNN-based methods, under a similar number of training parameters and FLOPs, L2S-Net achieved the optimal OA and recall rate. It is noteworthy that compared with visual methods, GNN-based methods usually have less trainable parameters and

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TABLE II

PERFORMANCE OF L2S-NET AND BASELINES ON THE ESADV2 DATASET. ♠ DENOTES CNN OR TRANSFORMER-BASED METHODS. ♦ INDICATES GNN-BASED METHODS. NOTABLY, PARAMS. DENOTES TRAINABLE PARAMETERS. THE *-L INDICATES THAT THE METHOD IS INCAPABLE OF HANDLING HIERARCHICAL GRAPHS, USING ONLY THE L-LEVEL GNN. THE **BEST** PERFORMANCE AND SECOND BEST RESULTS ARE HIGHLIGHTED

Model	Source	OA (%) ↑	Recall (%) ↑	Params. (M) ↓	FLOPs (G) ↓
♠ResNet-50	CVPR 2016	97.54	97.26	25.61	86.35
♠MobileNetV3	ICCV 2019	94.70	93.00	5.40	4.83
♠EfficientNet-B4	ICML 2019	95.93	95.82	19.34	1.43
♠ViT-B32	ICLR 2021	93.84	91.63	86.60	89.55
♠MobileViT-S	ICLR 2022	94.40	96.20	20.37	1.82
♠EfficientFormer-L1	NeurIPS 2022	91.57	92.78	11.39	27.37
♠MobileFormer	CVPR 2022	91.64	92.93	4.82	6.26
♠EMO-1M	ICCV 2023	90.56	91.25	1.19	5.53
♠EMO-5M	ICCV 2023	91.49	91.71	4.82	19.64
♦ HGP-SL-L	ArXiv 2019	93.91	94.00	0.14	8.1705
♦ HGP-SL-S	ArXiv 2019	88.15	88.27	0.13	8.1705
♦ Graph-Unets-L	ICML 2019	90.19	90.31	0.34	9.978
♦ Graph-Unets-S	ICML 2019	78.29	77.84	0.33	8.293
♦ MEWISPool-L	NeurIPS 2021	94.37	83.33	0.05	8.183
♦ MEWISPool-S	NeurIPS 2021	80.50	79.12	0.04	8.182
♦ HACT-Net	MIA 2022	95.91	95.33	0.92	8.204
♦ ViG-Ti	NeurIPS 2022	96.27	71.36	10.70	3.31
L2S-Net(Ours)	-	**98.20**	**98.19**	0.78	8.181

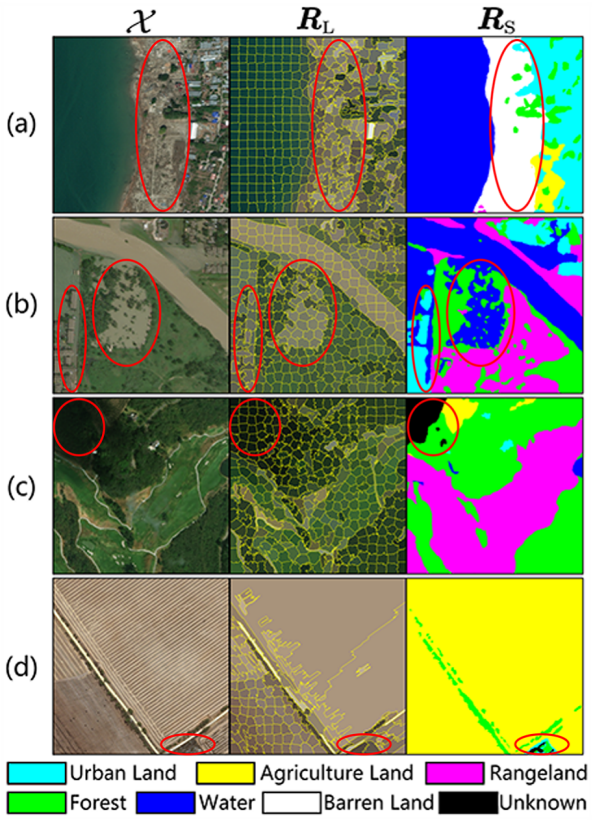

Fig. 10. Visualization of intermediate results of L2S-Net. The three columns denote $\mathcal{X}$, $\boldsymbol{R}_L$, and $\boldsymbol{R}_S$. (a) and (b) TP. (c) FP. (d) FN.

FLOPs. This is primarily due to the introduction of pretrained models and leveraging the characteristics of GNNs in data compression and lightweight structure design. Furthermore, in GNN-based methods, it is observed that methods using only L-level GNN usually outperform S-level one, mainly due to the loss of node information and the limited number of nodes in G_S. The L2S-Net, leveraging hierarchical graphs, excels in using local and semantic information, demonstrating superior performance. Similarly, HACT-Net, also using hierarchical graphs, significantly outperforms methods such as HGP-SL and MEWISPool, indirectly proving the effectiveness of our graph construction strategy. Notably, while ViG achieved comparable outstanding OA to L2S-Net, its recall rate is exceptionally low and the relatively large model size is disadvantageous for the target task.

2) Qualitative Results:

a) Qualitative analysis on ESADv2 dataset: To further analyze the advantages of L2S-Net in Earth's surface anomaly detection and evaluate the effectiveness of the hierarchical graph inspired by brain-inspired theories, we randomly selected a few intermediate results from the test samples as shown in Fig. 10. In the TP samples as shown in Fig. 10(a), the barren land (or beach) between urban land and ocean is commonly seen. However, the scattered houses, trees, and low-level visual features supplemented from $\boldsymbol{R}_L$ appeared after a tsunami guided the model for correct detection. In the flood anomaly scenario as Fig. 10(b), houses and trees surrounded by water are anomalous semantic relationships, which explicitly guides L2S-Net in anomaly detection, emphasizing the importance of integrating G_S.

We also observed that a few challenging scenarios will affect model robustness and misguide the model inference. As shown in Fig. 10(c), the FP sample exhibits normal land cover, yet the shadowy area in the upper left corner is incorrectly predicted as water body or unknown class in $\boldsymbol{R}_S$ while representing an anomaly of low-level information in $\boldsymbol{R}_L$, introducing noise into the model inference. Similarly, in the FN sample illustrated in Fig. 10(d), an inconspicuous anomalous area in the bottom right corner of vast, continuous agricultural land experiences a bushfire. These areas typically correspond to edge leaf nodes in graph, with a lower likelihood

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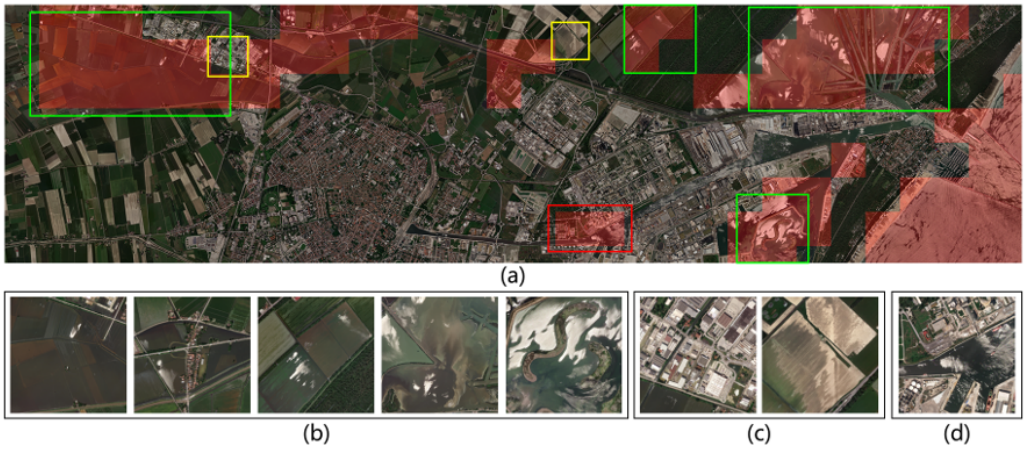

Fig. 11. Result of Emilia-Romagna flooding. (a) Area classified as Earth's surface anomaly is indicated by the red mask. (b) TP area is marked with green boxes, with its subregions zoomed in. (c) FN area is outlined in yellow boxes, with its subregions zoomed in. (d) FP area is depicted in red boxes, with its subregions zoomed in.

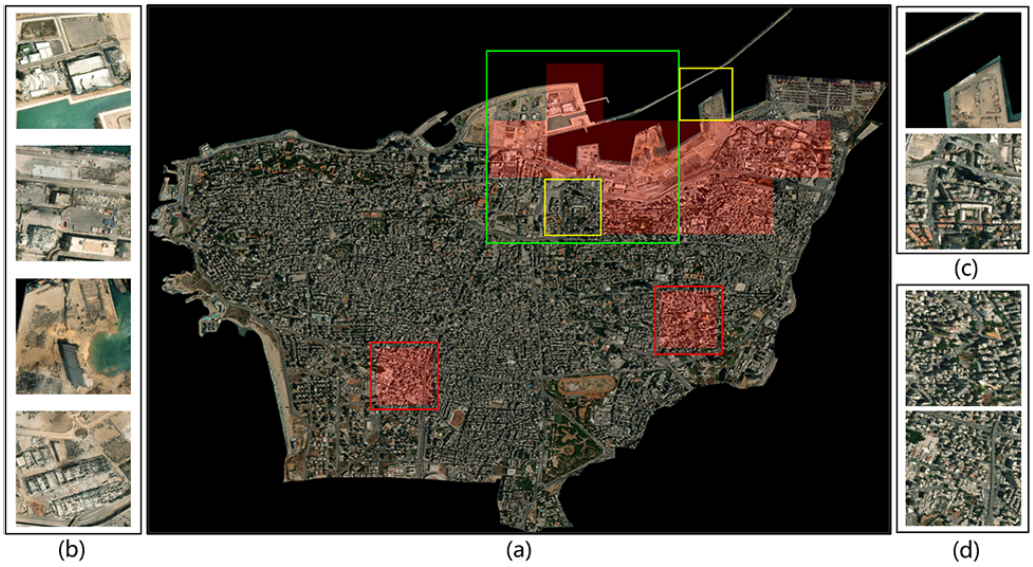

Fig. 12. Result of Beirut port explosion. (a) Area classified as Earth's surface anomaly is indicated by the red mask. (b) TP area is marked with green boxes, with its subregions zoomed in. (c) FN area is outlined in yellow boxes, with its subregions zoomed in. (d) FP area is depicted in red boxes, with its subregions zoomed in.

of spatial correlation with other ground objects and limited feature information available. The above scenario indicates that the limitations of the pretrained model have impacted the overall model performance. However, the majority of semantics and relationships still provide valuable information for correct detection. We believe that the performance of L2S-Net could be further improved in the future by replacing components with superior performance.

b) Qualitative analysis on real-world case study: Our objective is to achieve rapid and accurate Earth's surface anomaly detection in real-world scenarios, rather than merely exceling on datasets. Therefore, to validate the robustness and generalizability of L2S-Net, we conducted evaluations based on two real-world cases.

The Emilia-Romagna flooding affected a wide and dispersed area, as shown in Fig. 11. Fig. 11(b) provides a zoomed-in view of notable TP areas, such as farmlands completely submerged by floods and towns surrounded by water. These distinct anomalous relationships provide crucial support for the model's accurate detection. Fig. 11(c) shows a few FN areas, where inconspicuous and partially submerged farmlands lead to incorrect predictions. In the FP areas of Fig 11(d), even though the water bodies show normal distribution, the shimmering water surface introduces noise into G_{L}, deteriorating the model's decision.

As illustrated in Fig. 12(a), the Beirut port explosion primarily affected the surrounding areas of the explosion's epicenter. Fig. 12(b) shows that the affected core areas were accurately detected by the L2S-Net, with the TP regions displaying significant building damage or collapses. Conversely, Fig. 12(c) presents the FN regions, where the anomalous areas were too small or inconspicuous, leading to missed detection, probably due to image cropping and sliding window strategies. Fig. 12(d) displays the FP areas, characterized by dense and complex buildings that may introduce noise resulting in misclassification. Moreover, several areas marked as minor building damage in [8] were nearly invisible and were not detected by L2S-Net.

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TABLE III

ABLATION STUDY FOR ASSIGNMENT MATRIX. AM REPRESENTS THE ASSIGNMENT MATRIX. THE **BEST** RESULT IS HIGHLIGHTED

Model	OA (%)	Recall (%)
L2S-GNN-L	95.41	95.33
L2S-GNN-S	86.40	86.38
L2S-GNN w/o AM	96.75	96.73
L2S-GNN-S w AM	96.82	96.81
L2S-GNN w AM	**98.20**	**98.19**

TABLE IV

IMPACT OF DIFFERENT GNN LAYERS ON MODEL PERFORMANCE. THE **BEST** RESULT IS HIGHLIGHTED

GraphConv Type	Source	OA (%)	Recall (%)
GCN	ICLR 2017	96.25	96.19
GraphSAGE	NeurIPS 2017	95.21	95.37
GAT	ICLR 2018	97.01	96.86
EdgeConv	ECCV 2018	92.08	92.01
GIN	ICLR 2019	96.59	96.52
PNA	ICLR 2020	75.23	75.03
Max Relative GraphConv	TPAMI 2021	95.37	95.48
GATv2	ICLR 2020	**98.20**	**98.19**

TABLE V

IMPACT OF DIFFERENT FEATURE AGGREGATORS. THE **BEST** RESULT IS HIGHLIGHTED

Aggregator Type	OA (%)	Recall (%)
L2S-GNN	95.97	96.00
L2S-GNN w Summation	97.45	97.47
L2S-GNN w Mean	96.70	97.55
L2S-GNN w Concatenation	97.40	97.35
L2S-GNN w LSTM	**98.20**	**98.19**

In summary, through visual interpretation, L2S-Net has demonstrated satisfactory performance in real-world Earth's surface anomaly detection, particularly with a high recall rate. While there may be a certain FP rate and potentially FN due to the setting of the stride of the sliding window, it is acceptable for this task. We are able to efficiently achieve rapid and accurate detection of Earth's surface anomaly, thus saving valuable time for downstream tasks. Notably, L2S-Net has shown commendable generalizability in unseen classes of explosions.

3) Ablation Studies:

a) Assignment matrix for feature extraction: The assignment matrix describes the relationships between graphs at different levels and is the pivotal module in hierarchical graph. To verify the impact of the assignment matrix, we conducted a series of ablation studies. As shown in Table III, L2S-GNN-L and L2S-GNN-S, which only use G_{L} and G_{S} for anomaly detection, obtain inferior performance, especially when relying solely on G_{S}. Using both G_{L} and G_{S} and performing feature fusion based on AFM can increase the OA by about 1%. When the assignment matrix is used to reinitialize G_{L} and predictions are made solely using S-Level GNN, its performance is comparable to that of using AFM, highlighting the importance of supplement of local information. When all the components of L2S-GNN are used simultaneously, the model can achieve optimal performance, with OA and recall rates of 98.20% and 98.19%, respectively.

b) Impact of GNN layer: To validate the effectiveness of GATv2 as the core structure of L2S-GNN, we compared it with influential GNN layers in recent years, as shown in Table IV. The experimental results indicate that using GATv2 outperforms other GNN layers. This suggests that leveraging attention mechanisms to explore the diverse relationships among ground objects in Earth's surface anomaly detection can effectively significantly enhance the model inference capabilities, thereby improving its performance.

TABLE VI

IMPACT OF DIFFERENT FEATURE FUSION METHODS. THE **BEST** RESULT IS HIGHLIGHTED

Feature Fusion Method	OA (%)	Recall (%)
L2S-GNN-L	95.41	95.33
L2S-GNN-S	86.40	86.38
L2S-GNN w Summation	97.09	97.15
L2S-GNN w Mean	96.78	96.76
L2S-GNN w Concatenation	97.16	97.17
L2S-GNN w AFM	**98.20**	**98.19**

TABLE VII

IMPACT OF DIFFERENT NODE FEATURE REPRESENTATIONS. THE **BEST** RESULT IS HIGHLIGHTED

Num.	Node Feature Extraction Method		OA (%)	Recall (%)
	L-Level	S-Level		
1	RGB	Feature Map	76.90	76.70
2	RSP CNN	Feature Map	98.17	98.16
3	IMP CNN	Feature Map	**98.20**	**98.19**
4	IMP CNN	RGB	94.32	94.20
5	IMP CNN	IMP CNN	97.09	97.08
6	RSP CNN	RSP CNN	97.84	97.86

c) Feature aggregator for graph embedding: In the L2S-GNN, we leverage an LSTM module to capture the dependencies across graph embeddings from different layers, resulting in a final graph embedding. Table V highlights the significance of integrating features of different GNN layers, facilitating a more comprehensive graph representation. Compared with common feature aggregation methods (such as summation, mean, and concatenation), using LSTM in L2S-GNN demonstrates superior performance.

d) Impact of feature fusion: We conducted a series of experiments to validate the effectiveness of the AFM, as shown in Table VI. The experimental results demonstrate that the incorporation of the AFM module resulted in a significant improvement. In addition, we compared AFM with common feature fusion methods, such as summation, mean, and concatenation. It is evident that using the proposed AFM, we achieved optimal performance with only a marginal increase in model training parameters, resulting in an improvement of approximately 1%.

e) Impact of node feature representation: In the L2S-Net, hierarchical graphs are represented by nodes with attributes, graph topology, and assignment matrices, making the attributes of nodes critically important. In our work, we extract the node features of G_{L} using a pretrained CNN and the node features of G_{S} using feature maps extracted by the pretrained model represented as feature maps in Table VII). We conducted a series of experiments to compare various node feature extraction methods. As shown in Table VII, IMP refers to ImageNet pretrained ResNet-34, while RGB denotes the RGB values from the original VHR images. Given that Earth's

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surface anomaly detection is an RS task, we also explored models pretrained on RS datasets, specifically a ResNet-50 pretrained on the AID dataset, denoted as RSP CNN.

Experiments 1 and 4 demonstrate the negative impact of using RGB values as node features, particularly when used in G_{L}, significantly reducing useful low-level information and leading to inferior performance. Experiments 2, 3, and 6 show that using RSP CNN did not significantly improve performance compared with IMP CNN, possibly due to the smaller scale of the dataset used for pretraining the former and its potential lack of generalization to the ESADv2 dataset. Moreover, the RSP CNN used significantly increases the model parameters, which is not conducive to the target task. Experiments 3–5 indicate the advantages of using feature maps for representing node features of G_{S}, where leveraging feature maps closely related to semantic segmentation results enhances the expression of semantic information.

V. Conclusion

In this article, we propose a framework, called L2S-Net, for rapid and accurate Earth's anomaly detection. To alleviate the inherent response time delay in data collection and preprocessing, L2S-Net solely uses a single VHR satellite image as input for anomaly detection. Drawing from brain-inspired and graph theory, we introduce a hierarchical graph representation to integrate both local and semantic-level information of RS imagery. Furthermore, we propose the L2S-GNN to explicitly learn the relationships between ground objects, thereby enhancing model performance. We have instantiated L2S-Net and conducted extensive experiments on the proposed ESADv2 dataset and two real-world cases. The experimental results demonstrate that L2S-Net outperforms current state-of-the-art methods, showing exceptional robustness and generalization capabilities. We believe that the proposed L2S-Net and the benchmark dataset ESADv2 can further improve HADR-related work. In the future, we will further expand our research to include more comprehensive types of anomalies, more accurate and fine-grained results, open-world robust applications, and even on-orbit model deployment.

Acknowledgment

The authors would like to thank the authors of publicly available datasets xBD, Multi^{3}Net, Sichuan landslide and debrisflow dataset, GVLM, and Maxar Technologies for sharing satellite imagery of two real-world cases in this study. The numerical calculations were supported by the super-computing system at the Supercomputing Center, Wuhan University, Wuhan, China.

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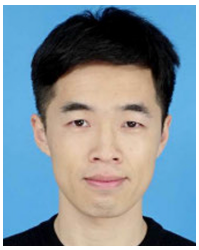

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